

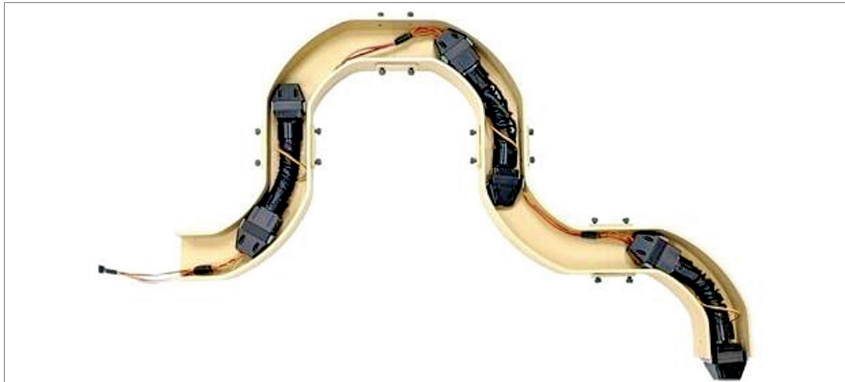


Fig. 2: Worm-like robot with artificial muscle created by Northwestern engineers (Credit: Northwestern University)

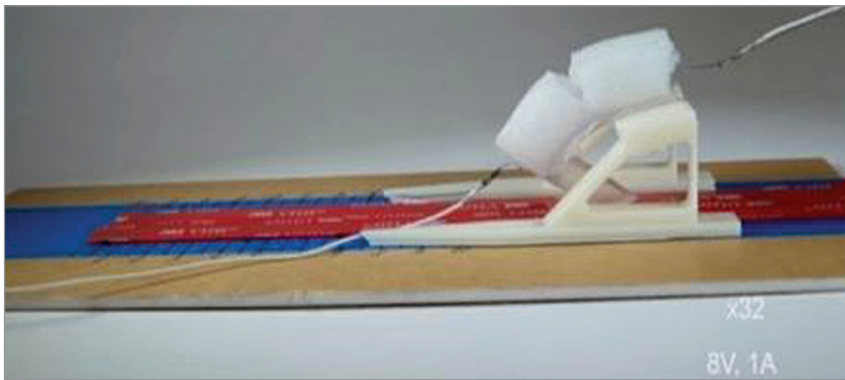


Fig. 3: Sleigh robot with a soft composite actuator (Credit: thenewsstck.io)

Transforming a mesh-like worm robot into an autonomous marvel

The Northwestern engineers' mesh worm can be transformed into an autonomous robot by integrating advanced sensors, actuators, and control systems.

First, the mesh worm can be equipped with cameras, infrared, and proximity sensors, enabling it to perceive its environment and make decisions based on the information gathered. These sensors allow the robot to navigate complex environments, avoid obstacles, and interact with objects.

Second, actuators such as motors and servos can be added to the mesh worm to enable it to move and manipulate objects. By controlling the movement of its segments, the

robot could crawl, slither, and even climb over obstacles.

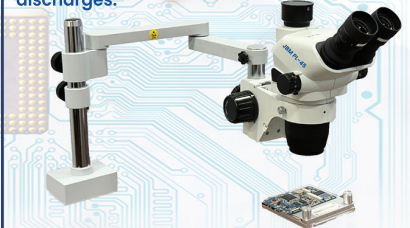
Third, a sophisticated control system could coordinate the actions of the sensors and actuators, allowing the robot to operate autonomously without human intervention. This control system could be programmed with algorithms enabling the robot to explore unknown environments, search for objects, and interact with its surroundings.

The rise of robotic muscle technology represents a significant milestone in robotics, opening up new possibilities for developing advanced robotic systems that can perform tasks beyond the capabilities of human muscles. As researchers continue to innovate and refine these technologies, such robots will likely be introduced in the future, undertaking increasingly complex functions. **EFY**



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